

Quantum Mechanics III

Homework 15

Seonwoo Park

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Topic

Blind-State Characterization

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Blind-State Characterization. State all assumptions and conventions.

$$R_{final} = R_{QDC26} \cup P_1^*, \dots, P_4^*$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$F(\rho_i^*, \hat{\rho}_i) > F_{registry}$$

3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.

$$release = hash(QDCdiff, states, regression, log)$$

Show enough intermediate work that another student can reproduce the calculation.